

ABHAY SHASHIDHARA

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PROFESSIONAL SUMMARY

M.S. Data Science candidate with 1+ year of experience in Data Science, AI engineering, and Data Engineering across healthcare, manufacturing, and pricing analytics. Skilled in building ELT workflows, API-based KPI ingestion pipelines, RAG systems, dashboard intelligence agents, and cloud-based machine learning solutions.

ACADEMIC QUALIFICATION

Master of Science in Data Science, University of Minnesota - Twin Cities, MN, USA | 2025-2027.

Relevant Coursework: Machine Learning Fundamentals, Principles of Database Systems, Introduction to Data Mining, Natural Language Processing, Spatial Data Science Research, Software Engineering.

Bachelor of Engineering, RV College of Engineering, Bengaluru, India | 2021–2025.

Minor in Artificial Intelligence, Indian Institute of Technology, Ropar | August 2024 - August 2025.

WORK EXPERIENCE

Radwell International, NJ, USA | Data Science Intern

June 2026–August 2026

- Building a **Databricks-native RAG assistant** using Knowledge Agent serving endpoints, semantic metadata and dashboard documentation to generate **Power BI summaries** for manufacturing and pricing analytics.
- Developing a **LightGBM-based dead-stock classification model** for inventory optimization across a **40M+** parts catalog by extracting **P21 database** features from stock movement, product metadata and pricing history.
- Automating Sales Operations reporting by developing an **OAuth-authenticated REST API** pipeline to extract **Genesys Platform** queue and agent metrics to replace manual workflows.
- Supporting competitor data collection for pricing analytics using **SerpAPI**-based web scraping workflows to capture product, pricing and availability signals from external sources.

BOCK AI, Bengaluru, India | AI and Automation Intern

December 2024 – June 2025

- Engineered **Python-based ETL pipelines** using **Pandas, JSON, and regex** to extract, clean, and standardize **10,000+ medical text samples** into structured datasets for LLM fine-tuning.
- Fine-tuned **DistilGPT-2** on domain-specific medical data using **gradient-based optimization**, validation monitoring, early stopping, and hyperparameter tuning to reduce the train-validation loss gap by **~18%**.
- Built and deployed a **scalable LLM fine-tuning pipeline on Google Cloud Vertex AI**, reducing experiment iteration time by **~30%** and enabling rapid retraining across medical sub-domains.
- Collaborated on a **medical text-to-video pipeline** prototype, leveraging **CogVideoX** and **transformer-based prompt encoders** to generate clinical instruction visualizations.

ACADEMIC PROJECTS

Evidence-Aware Hallucination Reduction for 3GPP RAN 🔗

- Built an **evidence-aware RAG system** for **3GPP TS 38.331 RAN** technical QA, tuning **BM25** retrieval, evidence-selection thresholds, and **Qwen2.5** model configurations to reduce hallucinations.
- Integrated a **DeBERTa-based NLI verification layer** using entailment and contradiction scoring to validate generated answers against retrieved TS 38.331 evidence.
- Implemented an **adaptive retrieval fallback for secondary verification** across **10 selected telecom evidence sources** to recover weak-evidence cases and reject unsupported responses before final output.
- Achieved **90% unanswerable rejection**, **10% hallucination rate**, and **0.642 answerability F1** under the evidence-aware RAG framework.

Detecting Co-Travelers with Temporal Detailed People Proximity Graphs 🔗

- Designed a novel **Temporal Detailed People Proximity Graph (TDPPG)** for **co-traveler detection** by weighting user-user edges using encounter frequency, shared venues, distinct days, and spatiotemporal proximity strength.
- Applied **PyMetis graph partitioning** to identify proximity-based user communities and group-level movement patterns from noisy location-based social network data.
- Conducted a **comparative evaluation** across **Gowalla** and **Brightkite** public **check-in datasets**, achieving **0.7588 ROC-AUC**, **0.4506 PR-AUC**, and **0.994 Precision@500** on Gowalla.
- Performed **ablation-style sensitivity analysis** on distance thresholds, time gaps, and event group sizes to optimize TDPPG performance and reduce noisy co-occurrence signals.

A Multimodal Retrieval-Augmented Diagnostic Framework for Parkinson's Disease Detection. 🔗

- Developed a **multimodal diagnostic system** combining structured patient metadata (e.g., age, comorbidities, symptom onset) and spiral test images for early-stage Parkinson's detection.
- Built a **RAG pipeline** using **OpenChat 3.5** and domain-specific transformers like **PubMedBERT** and **BioClinicalBERT** to generate clinician-style diagnostic responses.
- Fused **ResNet-18 CNN image features** with clinical metadata, outperforming image-only and metadata-only baselines by **6%+** while achieving **96.08% test accuracy** and **0.96 F1-score** across **255 test samples**.
- Prototyped the full system using **Gradio** to support real-time clinical inputs and generate interpretive text-based clinical summaries.

TECHNICAL SKILLS

- **Programming:** Python, C++, C, R.
 - **Data Engineering & Databases:** PySpark, Apache Spark, Delta Lake, Hadoop, PostgreSQL, SQL Server, MongoDB, ETL/ELT Pipelines, Medallion Architecture, API Data Ingestion, JSON Processing .
 - **Cloud & Data Platforms:** Databricks, Snowflake, Microsoft Fabric, Azure Data Factory, Google Cloud Platform.
 - **Developer Tools:** Git, GitHub, VS Code, Jupyter Notebook , SSMS
 - **Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face Transformers, SentenceTransformers, LangChain, OpenCV, FAISS, MLflow.
 - **Frameworks & APIs:** Flask, REST APIs, Streamlit, Gradio, Bootstrap.
 - **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Excel, Power BI, Tableau
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PUBLICATIONS

- De novo Design of Anti-COVID Drugs Using Machine Learning-Based Equivariant Diffusion Model Targeting the Spike Protein (Published: MDPI, 12 May 2023) [🔗](#)
 - Artificial Intelligence and Cloud Computing in Healthcare(Published: Springer Nature , 31 January 2026) [🔗](#)
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